



Xiaoxuan Zhang

AI & Quant Systems Engineer

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Focus Areas

- Time Series Foundation Models
- High-Availability Distributed Systems
- Crypto Trading Infrastructure
- Blockchain Protocol Development

Technical Stack

- Languages
 - C++, Golang, Python
- Paradigms
 - Event-Driven, Concurrency, Worker Pool, Producer-Consumer, Factory Pattern
- AI Engineering
 - PyTorch, Chronos2, W&B
- Infrastructure
 - Redis, MongoDB, Couchbase, MySQL, Kubernetes, NATS, Kafka
- Trading
 - TradingView, Pine Script

PROFILE

Software engineer with experience spanning distributed systems, blockchain infrastructure, and quantitative trading platforms. Currently focused on integrating transformer-based time-series models into real-world cryptocurrency trading pipelines, combining AI forecasting, statistical learning, and market infrastructure engineering.

EXPERIENCE

- 2024–now **AI-driven Crypto Forecasting Research** UNIVERSITY OF AMSTERDAM
Fine-tuning Chronos2 time-series foundation models for probabilistic cryptocurrency return forecasting, aided by classical statistical methods for model diagnostics and calibration.
- 2021–now **Crypto Trading Infrastructure** INDEPENDENT / OPEN SOURCE
Built real-time market data infrastructure: WebSocket stream subscriptions for candlestick data cached in Redis, with a dedicated persistence thread to MongoDB; RESTful API gap-filling ensures historical data completeness and consistency.
- 2021–2024 **Senior Software Engineer** MAVENIR
Engineered telecom-grade microservices (Golang/Kubernetes) for virtual network element management; message-driven via NATS (intra-cluster) and Kafka (inter-cluster). Led HA redesign: stateless services with Couchbase, shared-memory state sync, and load-balancer gateway for multi-instance resilience. Designed two-layer Goroutine concurrency middleware: channel-based worker pool with dynamic scaling at the bottom; closure-based async callback framework at the top.
- 2018–2020 **Blockchain Engineer** LAVA
Implemented consensus logic and storage optimisations for a Bitcoin-derived blockchain protocol in C++.

SELECTED PROJECTS

More at xiaoxuanzhang.com/#projects.

Chronos2 Crypto Forecasting

Fine-tuned foundation models for probabilistic return forecasting and uncertainty-aware signal generation.

Gofer — Goroutine Concurrency Middleware

Two-layer design: channel-based worker pool with dynamic scaling at the bottom; closure-based async callback framework at the top. Eliminates uncontrolled goroutine growth in high-concurrency telecom services. (Proprietary, Mavenir)

ChomoSyncer

A real-time exchange ingestion and persistence system serving AI-driven quantitative trading workflows.

EDUCATION

- 2024–2026 **M.Sc. Computational Science** UNIVERSITY OF AMSTERDAM
Focus: computational finance, parallel computing, AI-driven time-series forecasting, and quantitative trading systems.
- 2009–2015 **B.Eng. & M.Eng. Control Engineering** WUHAN UNIVERSITY
Strong foundation in mathematics, systems engineering, and software development.